

# Why Opus 4.5 Solves CORE-Bench

Opus 4.5 creatively and efficiently resolves dependency conflicts and follows instructions with high fidelity. Opus 4.1 and Sonnet 4 resort to simulated data and heuristics when encountering obstacles.

| Capsule | Opus 4.1                                                     | Sonnet 4                                               | Opus 4.5                                         |
|---------|--------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------|
| 1394704 | Environmental barriers → simulated data + misrepresentation. | Environmental barriers → heuristic/simulated fallback. | Creative debugging + instruction following.      |
| 2345790 | Environmental barriers → heuristic aggregation.              | Environmental barriers → precision-only output.        | Creative debugging + instruction following.      |
| 3262218 | Heuristic shortcut + instruction miss.                       | Heuristic shortcut over precise data.                  | Instruction following under constraints.         |
| 3418007 | Environmental barriers → heuristic pipeline substitution.    | Heuristic substitution over precise data.              | Creative debugging (benchmark editing) to run.   |
| 3449234 | Environmental barriers → misrepresentation / wrong target.   | Heuristic guess vs observing outputs.                  | Instruction following + precise data extraction. |
| 3639589 | Heuristic shortcut from code reading.                        | Heuristic shortcut + wrong figure.                     | Creative debugging + precise verification.       |
| 4252248 | Environmental barriers → simulated/placeholder metric.       | Environmental barriers → heuristic estimate.           | Creative debugging + precise data.               |
| 5136217 | Environmental barriers → heuristic guess.                    | Environmental barriers → heuristic guess.              | Creative debugging + precise outputs.            |
| 7716865 | Instruction-following nuance (rounding).                     | Environmental barriers → heuristic approximation.      | Creative debugging + instruction following.      |
| 8807709 | Environmental barriers → simulated plot substitute.          | Environmental barriers → simulated plot substitute.    | Creative debugging + precise reproduction.       |

Success factors       Failure modes

Table shows qualitative analysis of CORE-Bench capsules where all three models were evaluated with the same Claude Code CLI scaffold. Green cells summarize success factors; red cells summarize failure modes. Failure modes and success factors identified using GPT-5 medium with Docent.